

Spring Boot Practice Assignment

Dynamic Object Creation with ObjectProvider

Learning Goals

By completing this assignment, students should be able to:

- Understand prototype vs singleton scope
 - Use ObjectProvider for dynamic object creation
 - Pass arguments for constructor-based creation
 - Dynamically assign dependencies at runtime
 - Use Map injection for selecting implementations
 - Explain how Spring manages object creation
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Part 1 — System Setup (Conceptual)

You are building a **University Learning System**.

The system contains:

1. Course

- Represents a subject a student can take
- Examples:
 - Java
 - Python
 - Web Development

2. Student

- Has:
 - name
 - GPA
 - assigned Course
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Task 1:

Design the system so that:

- Course has multiple implementations

- Each Course is managed by Spring
 - Student is also managed by Spring
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Task 2:

Decide:

- What scope should Student have? (singleton or prototype?)
 - Explain your reasoning
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Part 2 — Using ObjectProvider

Task 3:

Explain how you would use ObjectProvider to:

- Create a new Student each time it is requested
 - Ensure that two students are not the same object
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Task 4:

Describe how ObjectProvider differs from using `new` keyword



Part 3 — Constructor-Based Creation

Task 5:

You want to create a Student with:

- name
- GPA

Explain:

- How ObjectProvider allows passing values during creation

- Who decides which constructor is used
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Task 6:

What happens if:

- you pass arguments that do not match any constructor?

Explain expected behavior.



Part 4 — Dynamic Dependency Assignment

Task 7:

You want each Student to be assigned a Course dynamically at runtime.

Explain:

- Why you should NOT hardcode Course inside Student
 - How you would assign Course after creating Student
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Task 8:

Explain how Spring can inject all Course implementations into a Map.

Then describe:

- How you would select a Course using a string (e.g., "java", "python")
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